Project Learning from Data: Ethics

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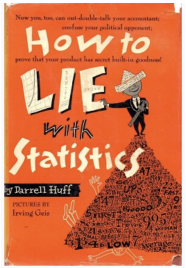
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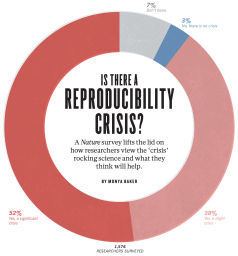
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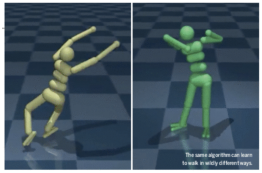


Introduction

Introduction







COMPUTER SCIENCE

Artificial intelligence faces reproducibility crisis

Unpublished code and sensitivity to training conditions make many claims hard to verify

Baker, M. (2016, Nature, 533:26): main reasons for reproducibility crisis:

- bad experimental / study design
- bad statistics
- poor mentorship

Introduction

Some other reasons for the reproducibility crisis:

- publication bias (journal more likely to publish significant results)
- publish or perish

Publishing, reporting and eventually using false results may harm people and society:

- a drug that is approved by the FDA based on wrong data analysis results may cause health problems or even death to the patients
- a wrong environmental risk assessment may result in the advice for continuation of discharge of industrial waste water: risk for public health and the environment
- a wrong assessment of financial risks may bring companies to bankruptcy

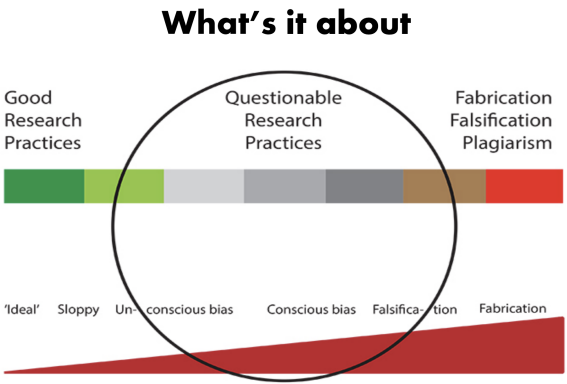
Introduction

Statisticians and data scientists

- bear an important responsibility to science and to society
- should know the applicable ethical principles and behave accordingly (scientific integrity is here considered to be part of ethics)
- should know the consequences and perspectives of all stakeholders involved, as well as societal relevance of their work

Introduction

Scientific Integrity (not only for statisticians / data scientists)



Source: Adapted from a presentation by Daniel Fanelli – by VIB

Introduction

Aim of this lecture:

- Explain the basic ethical principles for the profession of statistician / data scientist (we will use the guidelines of the Royal Statistical Society and the American Statistical Association)
- Illustrate these principles based on some examples

Examples

Examples: the Benveniste affair

The Benveniste affair

- A French immunologist, Jacques Benveniste claimed that water could "remember" substances that had previously been dissolved in it, even after those substances were diluted to the point where no molecules remained. This phenomenon was presented as a potential scientific basis for homeopathy.
- Benveniste and his team published a paper in the prestigious journal Nature claiming that highly diluted solutions of antibodies could still trigger a biological response, suggesting that water might retain some sort of "memory" of the substances it had contained. This notion was at odds with established principles of chemistry and physics.
- After the publication, Nature took the unprecedented step of sending an investigative team to Benveniste's lab to reproduce the results (the editor of Nature, a magician, and a scientific fraud investigator).
  The team visited Benveniste's lab and oversaw seven attempts to replicate the results:
  - first four attempts were faverable to Beneviste
  - The last three attempts were double blinded and contradicted Beneviste's claims
- The team concluded that unintentional bias was the culprit.

Examples: the Mark Regnerus Study

The Mark Regnerus Study

- Sociologist Mark Regnerus, a professor at the University of Texas at Austin, published a study in Social Science Research in 2012 titled "New Family Structures Study" (NFSS).
- He concluded: children of parents who had same-sex relationships fared worse on various social, emotional, and economic measures compared to children raised by heterosexual parents.
- The findings were quickly picked up by media outlets and used in public debates about same-sex marriage and parenting.

Examples: the Mark Regnerus Study

- His study grouped together people who had lived with a parent who had ever had a same-sex relationship, regardless of whether the child was raised in a same-sex household. Many of the children categorized as being raised by "same-sex parents" had experienced family disruption, divorce, or other adverse situations unrelated to same-sex parenting.
- The vast majority of the children in the "same-sex parent" group were raised by different-sex parents for most of their lives, which distorted the results and led to unfair comparisons.
- He used the wrong statistical data analysis method, ignoring confouders
- The study was funded by conservative organizations known for opposing same-sex marriage (e.g. the Witherspoon Institute and the Bradley Foundation). This raises concerns about potential bias, as the funding sources may have influenced the design of the study and the interpretation of its results.
  Mark Regnerus did not disclose the funding organisations.
- The study was rushed through the peer review process and published within a remarkably short time, raising concerns about whether it had undergone rigorous scrutiny. Several scholars pointed out serious methodological flaws after the study was published, leading to a re-evaluation of the journal's peer review process.

Examples: the Mark Regnerus Study

Consequences:

- Widespread criticism: The study faced harsh criticism from other social scientists for its methodological flaws. The American Sociological Association filed an amicus brief in the U.S. Supreme Court, arguing that Regnerus' study was flawed and did not provide credible evidence about the effects of same-sex parenting.
- Retraction Calls: Some researchers called for the retraction of the study due to its flawed methodology and misleading conclusions. Although the paper was not retracted, the controversy prompted discussions about the importance of rigorous statistical analysis and ethical standards in research.
- Reputational Damage: The reputational damage affected both Regnerus and the journal Social Science Research.
- Impact on Public Policy: Despite its flaws, the study was cited in court cases and political debates over same-sex marriage, contributing to the legal arguments against it in the U.S. before the Supreme Court's decision in Obergefell v. Hodges (2015), which legalized same-sex marriage nationwide.

Examples: Andrew Wakefield's Vaccine Study

Andrew Wakefield's Vaccine Study:

- n 1998, Andrew Wakefield, a British physician, published a paper in The Lancet suggesting a link between the MMR (measles, mumps, and rubella) vaccine and autism. The paper described a small study of 12 children and claimed that the onset of autism-like symptoms coincided with the administration of the MMR vaccine.
  Ethical violations:
  - Small Sample Size: The study used only 12 children, a sample size far too small to draw meaningful conclusions about the general population.
  - Selective Reporting: Wakefield was accused of cherry-picking data and selectively reporting only those cases that fit his hypothesis. Other data points that contradicted his claims were not included.
  - Conflicts of Interest: Wakefield had significant financial conflicts of interest, as he was involved in a lawsuit against vaccine manufacturers and stood to benefit from a negative result.
  - Falsified Data: Further investigations found that some of the data on the children's medical histories had been manipulated or falsified to fit the hypothesis.

Examples: Andrew Wakefield's Vaccine Study

Consequences:

- The paper was retracted by The Lancet in 2010, and Wakefield lost his medical license
- Damage: Wakefield's fraudulent claims led to a significant decline in vaccination rates and contributed to the rise of the modern anti-vaccine movement, resulting in outbreaks of preventable diseases.

Examples: the Diederik Stapel Scandal

The Diederik Stapel Scandal

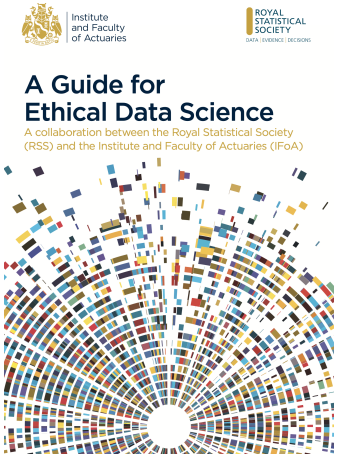
- Diederik Stapel was a prominent Dutch academic in social psychology, known for his research on human behavior, prejudice, and stereotypes. Over the course of his career, Stapel published numerous influential papers, many of which were later found to be based on fabricated data.
  The scandal came to light in 2011 when junior researchers in his lab began to notice irregularities in his work and raised concerns.
- How did he committed the fraud:
  - Fabrication of Entire Data Sets: instead of conducting real experiments, Stapel fabricated entire data sets to support his hypotheses.
  - No Actual Experiments: in many cases, Stapel did not conduct any experiments at all. He simply invented data that aligned with his theories.
  - Isolation from Collaborators: Stapel frequently controlled the entire research process himself, which allowed him to manipulate data without others noticing. He kept students and collaborators away from the raw data.

Examples: the Diederik Stapel Scandal

- How was it discovered?
  In 2011, three of Stapel's Ph.D. students noticed inconsistencies in his work and reported their concerns. This led to a formal investigation by universities where Stapel had worked. The investigation revealed that Stapel had falsified data in 55 papers over a period of more than a decade. 30 PhD dissertations were based on his fabricated research.
- Consequences:
  - Academic Fallout: Stapel was dismissed from his position at Tilburg University, and many of his papers were retracted. His fraudulent research undermined the credibility of social psychology and damaged public trust in the field.
  - Retraction of Publications: Journals that had published his fraudulent studies retracted his papers. The scope was extensive, and entire lines of research had to be reconsidered.
  - Impact on Students and Collaborators: Many of Stapel's PhD students and collaborators were negatively affected by the scandal, as their careers were based on work that had been falsified.
  - Impact on Social Sciences: The Stapel case sent shockwaves through the social sciences and prompted a deeper examination of research practices, especially around data transparency and replication. It led to increased scrutiny of published work and calls for more robust methods to detect fraud, such as preregistration of studies and open data practices.

RSS Guidelines

RSS Guidelines



RSS Guidelines

Theme 1: Seek to enhance the value of data science for society

- As the impact that data science can have on society could be significant, an important ethical consideration is what the potential implications could be on society as whole.
- A common theme within ethical frameworks discussing data science and AI is for practitioners to attempt to seek outcomes within their work which support the improvement of public wellbeing.

Theme 1: Seek to enhance the value of data science for society

What do we mean with "society"? It can e.g. mean

- education
- environment
- equality and inclusion
- fairness
- health and hunger
- security and justice
- economic empowerment
- infrastructure
- cyber risk and national security
- ...

Theme 1: Seek to enhance the value of data science for society

Some questions you can ask yourself / some suggestions:

- Who are the relevant stakeholders? What is their role (e.g. in society)?
- What impact may your conclusions have for these stakeholders and for society?
- Who or what can be effected by the outcomes of your work?
- Choose or seek work that is considered to have a benefit to the public/society
- Invite peers to review and discuss your work

Theme 2: Avoid harm

- Data science has the potential to cause harm and this ethical consideration therefore focuses on how practitioners can avoid this by working in a manner that respects the privacy, equality and autonomy of individuals and groups, and speaking up about potential harm or ethical violations.
- Statisticians and data scientists may be subject to legal and regulatory obligations in relation to the privacy of individuals, relevant to the jurisdiction in which they are working, as well as regulatory obligations to speak up about harm or violations of legal requirements.
- This can also be applied to work relating to businesses, animals or the environment, with consideration of commercial rights, animal welfare and the protection of environmental resources.

Theme 2: Avoid harm

Examples of harm:

- harm to health (physical / mental)
- financial harm
- harm to the reputation or privacy of persons, companies, institutions, ...
- harm to the environment
- harm to property or infrastructure
- ...

Theme 2: Avoid harm

Some questions you can ask yourself / some suggestions:

- Where does the data come from? How was it collected?
- Investigate presence of informed consent (if required) or other legal requirements
- Consider and respect the privacy, dignity and fair treatment of individuals
- Follow good data handling practices, including data security, and considering ways to reduce the need for collecting, storing and utilising personally identifiable information
- Consider the potential impact of linking different data sources
- Be transparent and provide evidence of privacy considerations to the public and/or regulators

Theme 2: Avoid harm

Some questions you can ask yourself / some suggestions:

- Use ethical assessment checkpoints
- Educate others about what data science can and cannot do
- Fully understand error, uncertainty, bias and risks associated with the methods used
- Evaluate the energy-cost of storing and processing large volumes of data, and considering different options such as cloud computing, aggregating data and regularly reviewing the need for data
- Invite peers to review and discuss your work

RSS Guidelines

Theme 3: Apply and maintain professional competence

- This ethical principle expects data science practitioners to apply best practice and comply with all relevant legal and regulatory requirements, as well as applicable professional body codes.
- Professional competence involves fully understanding the sources of error and bias in data, using 'clean' data (eg edited for missing, inconsistent or erroneous values), and supporting work with robust statistical and algorithmic methods that are appropriate to the question being asked.
- Practitioners can also thoroughly assess and balance the benefits of the work versus the risks posed by it, and keep models under regular review.

Theme 3: Apply and maintain professional competence

Some questions you can ask yourself / some suggestions:

- Comply with relevant professional and regulatory practices:
  - What are the relevant professional and legal/regulatory requirements?
  - Ensure that the work complies with these requirements and justify this to relevant authorities
  - Communicate these appropriately to relevant stakeholders
- Understand the limitations of your work (e.g. methods/models) and communicate this to your colleagues / superiors
- Follow best practices within your discipline / field of application

Theme 3: Apply and maintain professional competence

Some questions you can ask yourself / some suggestions:

- Have regular reviews of your methods/models.
- Share appropriate code, diagnostics and results with colleagues/customers to ensure they sufficiently understand the methods/models
- Publish results publicly where possible
- Invite peers and subject matter experts to review and discuss your work

Theme 4: Seek to preserve or increase trustworthiness

- The public's trust and confidence in the work of data scientists can be affected by the way ethical principles are applied. Practitioners can help to increase the trustworthiness of their work by considering ethical principles throughout all stages of a project.
- This is another reoccurring theme that encourages practitioners to be transparent and honest when communicating about the way data is used. Transparency can include fully explaining how algorithms are being used, if and why any decisions have been delegated, and being open about the risks and biases.
- Engaging widely with a diverse range of stakeholders and considering public perceptions both from the outset, and throughout projects, can help to build trustworthiness and ensure all potential biases are understood.

Theme 4: Seek to preserve or increase trustworthiness

Some questions you can ask yourself / some suggestions:

- Document the performance of different methods/models tested
- Motivate the choices you made along the way
- Communicate the ways in which data is used and the benefits to customers/ wider public
- Take accountability for involvement in the work
- Publish and share methods with relevant internal stakeholders, customers and the wider public, clearly referencing limitations, bias and uncertainty
- Be honest and objective in communications

Theme 5: Maintain accountability and oversight

- Another key issue in data ethics around automation and AI is the question of how practitioners maintain human accountability and oversight within their work.
- Being accountable can include being mindful of how and when to delegate any decision making to systems, and having governance in place to ensure systems deliver the intended objectives.
- When deciding to delegate any decision making, it would be useful to fully understand and explain the potential implications of doing so, as the work could lead to introducing advanced AI systems which do not have adequate governance.
- Practitioners should note that delegating any decisions to these systems does not remove any of their individual responsibilities.

Theme 5: Maintain accountability and oversight

Some questions you can ask yourself / some suggestions:

- Maintain human oversight of automated solutions:
  - Implement model governance by deciding how to monitor the method/model over time, such as setting review points and sign-off
  - Agree a review process
- Add clear constraints within systems and corresponding documentation to control and communicate what systems can and cannot do
- Have ethics committees where appropriate
- Have clear processes for accountability so that stakeholders have a clear route to any redress

ASA Guideline

ASA Guidelines

Approved by ASA Board of Directors
April 14, 2018

Ethical Guidelines for Statistical Practice

Prepared by the Committee on Professional Ethics
of the American Statistical Association

Purpose of the Guidelines

ASA Guidelines

Principles:

- Professional Integrity and Accountability
- Integrity of data and methods
- Responsibilities to Science/Public/Funder/Client
- Responsibilities to Research Subjects
- Responsibilities to Research Team Colleagues
- Responsibilities to Other Statisticians or Statistics Practitioners
- Responsibilities Regarding Allegations of Misconduct
- Responsibilities of Employers, Including Organizations, Individuals, Attorneys, or Other Clients Employing Statistical Practitioners

ASA Guidelines

We mention here some further items that are included in the ASA guidelines, but not explicitly in the RSS guidelines.
Data:

- Report the sources and assessed adequacy of the data, accounts for all data considered in a study, and explain the sample(s) actually used
- Clearly and fully report the steps taken to preserve data integrity
- Share the data used in the analyses whenever possible/allowable
- Strive to avoid the use of excessive or inadequate numbers of research subjects by making informed recommendations for study size

ASA Guidelines

Use of statistical methods:

- Select methods that are appropriate for answering the research questions.
  Do not select methods because of a desirable outcome of the data analyses with these methods.
- Acknowledge statistical and substantive assumptions made in the execution and interpretation of any analysis
- Where appropriate, address potential confounding variables not included in the study.
- Acknowledge data editing and data wrangling procedures, including any imputation and missing data mechanisms
- Report the limitations of statistical inference and possible sources of error

ASA Guidelines

Communication / publication:

- Respect and acknowledge the contributions and intellectual property of others
- Disclose conflicts of interest
- In publications / communication, identify who is responsible for the statistical work
- In publications / communications, identify the ultimate financial sponsor of the study, the stated purpose, and the intended use of the study results
- When reporting analyses of data that may not be representative of a defined population, include appropriate disclaimers and, if used, appropriate weighting
- Promote sharing of data and methods as much as possible and as appropriate without compromising propriety

ASA Guidelines

Some final items:

- Recognize any statistical descriptions of groups may carry risks of stereotypes and stigmatization.
  Statisticians should be sensitive to the manner in which information is framed to avoid disproportionate harm to vulnerable groups
- Recognize other professions have standards and obligations, research practices and standards can differ across disciplines, and statisticians do not have obligations to standards of other professions that conflict with these guidelines
- Support sound statistical analysis and expose incompetent or corrupt statistical practice.

Ethics Checklists

Ethics checklists

Many professional organisations have ethics checklists (e.g. RSS).



Project planning

To embed ethics when defining, scoping and initiating projects, you can:

- ☐ Complete an ethics assessment including:
  - Is it in the public interest, and how might value be distributed fairly?
  - Can data be ethically sourced?
  - Are there risks (privacy, harm, fairness) for individuals, groups, businesses, environment?
- ☐ Engage with the public/stakeholders to gather perceptions
- ☐ Seek early feedback from domain experts
- ☐ Define the governance for the project, including data security and handling

Data management

To embed ethics when acquiring, storing and preparing data for analysis, you can:

- ☐ Fully understand the consents and legal uses of the data
- ☐ Have data security measures in place, and ensure staff are trained in data handling
- ☐ Consider impacts of data processes to privacy, bias and error; including linking data sources, estimating demographic or other omitted information

Professional competence might include:

- ☐ Thorough quality checks
- ☐ Applying best practices for data cleaning (errors, inconsistency, missing values)
- ☐ Detecting and mitigating sources of bias

Analysis and development

To embed ethics when analysing data, building models and designing systems, you can:

- ☐ Apply consents and permitted uses, professional and regulatory requirements
- ☐ Monitor risks identified at planning, assess for additional risks (harm, bias, error, privacy)

Professional competence might include:

- ☐ Best practices, analytical rigour, quality assurance of methods, including peer review
- ☐ Minimising complexity in models and algorithms, validating thoroughly
- ☐ Fully explaining outcomes and uncertainty when making recommendations

Implementation and delivery

To embed ethics when operationalising and deploying models and systems, you can:

- ☐ Be transparent about when, how and why decisions have been delegated
- ☐ Share methods, results/limitations and code with relevant parties; invite feedback

Professional competence might include:

- ☐ Applying best practice in anonymisation before sharing data or disseminating outputs
- ☐ Adequate quality assurance of systems
- ☐ Regular reviews of models, with risk assessments

Communication and oversight

To embed ethics in communications:

- ☐ Communicate with stakeholders throughout projects and help any non-technical individuals understand the technical and ethical issues
- ☐ Be honest and objective when communicating about work

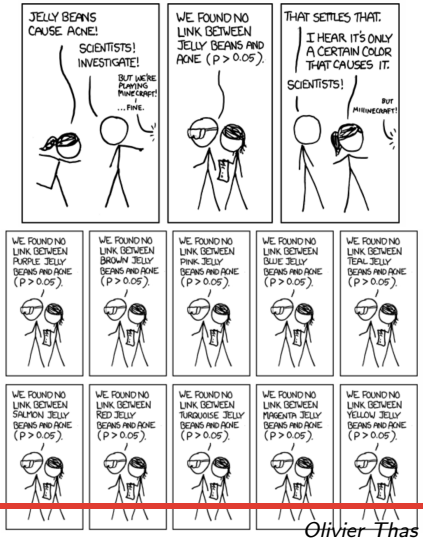
Leaders can:

- ☐ Ensure the organisation has and understands ethical principles and policies
- ☐ Engage with ethical bodies, privacy groups, stakeholder groups, the public

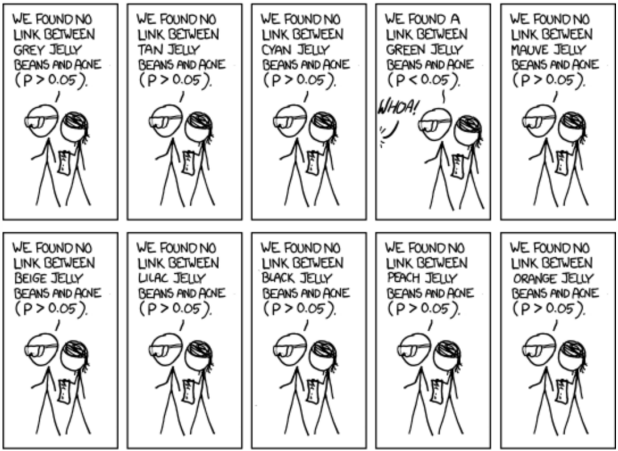
Examples of Misconduct

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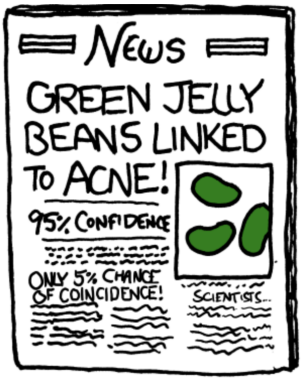
P-hacking or data dredging



Examples of Misconduct: p-hacking



Examples of Misconduct: p-hacking



Solutions:

- adjustment for multiple testing
- starting from a specific hypothesis and only report results for this analysis
- correct communication about data analysis procedure

Example of p-hacking

Brian Wansink's research focused on food choices and eating behavior, and he published widely cited studies on topics like portion sizes and plate colors affecting how much people eat.
Ethical violations:

- P-Hacking and Data Manipulation: Wansink was found to have engaged in p-hacking, using multiple analyses and data dredging to find significant results after data collection. He encouraged his lab members to try various methods and interpretations of data until statistically significant results emerged, even if they were not meaningful.
- Reproducibility Concerns: Several of Wansink's findings were challenged for being irreproducible, and later investigations showed that the data had been manipulated to achieve desired outcomes.

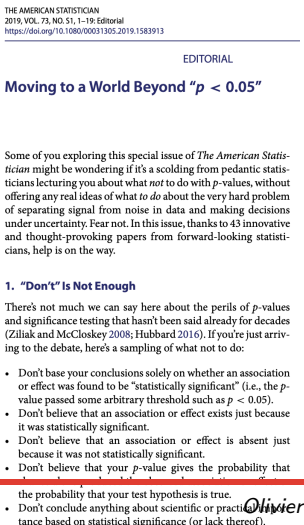
Consequences:

- Wansink's papers were widely retracted (over 15 as of 2018), and he resigned from his position at Cornell University. His case brought widespread attention to the issues of reproducibility, p-hacking, and unethical statistical analysis in behavioral science.
- Misinforming public policy and public health decisions, eventually putting health of people at risk.

Examples of Misconduct

Misuse of p-values

Wasserman et al. (2019, The American Statistician, 73):



THE AMERICAN STATISTICIAN
2019, VOL. 73, NO. S1, 1–19: Editorial
https://doi.org/10.1080/00031305.2019.1583913

EDITORIAL

Moving to a World Beyond "$p < 0.05$"

Some of you exploring this special issue of *The American Statistician* might be wondering if it's a scolding from pedantic statisticians lecturing you about what *not* to do with p-values, without offering any real ideas of what *to do* about the very hard problem of separating signal from noise in data and making decisions under uncertainty. Fear not. In this issue, thanks to 43 innovative and thought-provoking papers from forward-looking statisticians, help is on the way.

1. "Don't" Is Not Enough

There's not much we can say here about the perils of p-values and significance testing that hasn't been said already for decades (Ziliak and McCloskey 2008; Hubbard 2016). If you're just arriving to the debate, here's a sampling of what not to do:

- Don't base your conclusions solely on whether an association or effect was found to be "statistically significant" (i.e., the p-value passed some arbitrary threshold such as $p < 0.05$).
- Don't believe that an association or effect exists just because it was statistically significant.
- Don't believe that an association or effect is absent just because it was not statistically significant.
- Don't believe that your p-value gives the probability that [illegible] the probability that your test hypothesis is true.
- Don't conclude anything about scientific or practical importance based on statistical significance (or lack thereof).

Examples of Misconduct: Misuse of p-values

Consider the following two examples. In each example a new drug for body temperature reduction is compared to placebo:

1. Drug A:
   $\hat{\Delta} = 0.247$ °C with 95% CI: $[0.06$ to $0.43]$ °C; $p = 0.009$
2. Drug B:
   $\hat{\Delta} = 0.537$ °C with 95% CI: $[-0.06$ to $1.13]$ °C; $p = 0.074$

What drug seems worth of further investigation?
Note: in the study with drug A there were 15 subjects per group, and in the study with drug B there were 200 subject per group.

Examples of Misconduct: Selection Bias

The USA 1936 elections: Roosevelt versus Landon

- The Literary Digest predicted that Landon would win (55% vs. 41%)
  (based on 25% of the electorate $\approx$ 10,000,000)
- Mr. Gallup predicted that Roosevelt would win
  (based on 50,000 respondents)
- Real outcome of the elections:
  Roosevelt won with 61% versus 37%

Examples of Misconduct: Selection Bias

What happened?

- The Literary Digest mailed out forms to a list compiled from automobile registrations and telephone directories. Their sample was therefore not representative of population.
- Landon supporters turned out to be more likely to mail back their answers. The Literary Digest could not obtain results for the complete sample, hence missing data.

Food for Thought: a case study

Food for Thought: a case study

Consider the following setting:

- You work for a young start-up company in the biopharma industry
- The company is still not profitable and runs on money from investors and a seed fund
- The company aims to develop a diagnostic test for early stage colorectal cancer (CRC) based on a stool sample. They hope to find good biomarkers in
  - circulating free DNA (fcDNA) – circulating tumor DNA (ctDNA): SNPs in the tumor DNA
  - microbiome: relative abundances of thousands of microorganisms
- You, together with a direct colleague, work as statisticians / data scientists
- When the diagnostic kit can be realised and marketed, you will receive a bonus.

Food for Thought: a case study

What would you do in the following situations?

- You receive a dataset from a retrospective study. The dataset contains names of the subjects, as well as information on their medical history.
  Privacy? GDPR legislation?
- You check whether all participants have signed *informed consent*, but for 5 participants the document is missing.
  Informed consent is a legal requirement.
- Your colleague-statistician notices that there are missing values in data dataset and decides to impute them. He/she does not tell you, but you notice it because you can compare the data with the previous version of the dataset.

Food for Thought: a case study

What would you do in the following situations?

- The dataset has only 30 subjects.
- You identify a few outliers (2 patients are much older than the others, and for 2 other patients the pathogen *Clostridium difficile* was detected).
- The data needs preprocessing (normalisation, batch correction, ...). Many preprocessing methods are available. There is one method that gives good downstream results, but from literature you know that it is not a good preprocessing method.
- The development of diagnostic test requires building a good prediction model for CRC, and this requires setting a threshold that gives a good compromise between the sensitivity and specificity of the test. You are involved in the discussions. What are your considerations?

Food for Thought: a case study

- The development of the diagnostic method is finished, and it will be marketed soon. You are invited to give presentations to
  - a committee advising the authorities/government/public health insurance
  - organisation of CRC patients
  - hospitals and oncologists
- The final prediction model is based on data from only two hospitals.
- Your colleague starts with a new project in the company, and he re-uses the same dataset.

Conclusions

Conclusions

- The impact of your work on stakeholders and society may be large
- You need to be aware of the ethical principles and behave accordingly
- A few keywords:
  - Transparancy
  - Communication / documentation / openess
  - Share data/methods/insights and invite peers to review
  - Respect privacy and integrity of individuals / groups of people / organisations
  - Responsibility and accountability

Conclusions

Two final thoughts:

- Honesty is the best policy!
- It takes courage!